Effects of a Virtual Reality-Based Cycling Exercise Program on Engagement and Gait Outcomes in Older Adults

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Abstract

The ageing population faces increasing challenges in maintaining mobility and independence. Virtual reality (VR) may enhance exercise by increasing engagement and influencing perceived effort, emotional state, and motivation. This study compared a six-week cycling program performed with and without VR in very old adults. Thirty participants completed the intervention. Both groups performed 20 minutes of cycling at a constant speed of 6 km/h, three times per week. Physical performance was assessed using the Timed Up and Go test. Perceived effort was measured with the Borg scale, emotional state with the EVEA questionnaire, and motivation in the VR group with the Intrinsic Motivation Inventory. Descriptive statistics and independent t-tests were applied ($p < 0.05$). No statistically significant differences were observed between groups in physical performance. Perceived effort decreased in both groups, with a greater reduction in the VR group. Emotional state remained stable, with low levels of anxiety, depression, and hostility, while joy tended to be higher in the VR group. Motivation was maintained throughout the intervention. These findings suggest that VR may improve the exercise experience in very old adults and support its use as a complementary strategy in exercise programs for this population.

1. Introduction

The ageing of the population represents a major challenge for healthcare systems, particularly in maintaining mobility, independence, and quality of life in older adults. Regular physical exercise is essential to preserve physical function and reduce the risk of falls; however, adherence to exercise and rehabilitation programs in elderly populations is often limited due to low motivation, fatigue, or the monotony of conventional training.

In this context, technological solutions have been increasingly explored to support physical activity in older adults. VR has emerged as a promising tool to enhance exercise engagement by providing immersive and motivating environments [1]. Beyond potential physical benefits, VR may influence perceived effort, emotional state, and motivation, which are key factors for sustaining participation in exercise programs over time [2].

Previous studies have investigated VR-based cycling or exercise in older adults, but they have typically examined different outcomes separately rather than providing an integrated evaluation. Some studies focused on usability or user acceptance of VR systems, while others assessed physical performance alone [1]-[3]. None of these approaches combined physical function, perceived effort, emotional responses, and motivation within the same intervention and study design. In addition, most existing research has been conducted in relatively healthy and autonomous older adults, whereas very old populations remain underrepresented. The present study addresses these limitations by evaluating a VR-assisted cycling program in a very old population and integrating physical, perceptual, emotional, and motivational measures over a six-week intervention, providing a more complete understanding of the impact of VR-based exercise in advanced age. Accordingly, the aim of this study is to compare a six-week cycling program performed with and without VR in a population of very old adults.

2. Methodology

2.1. Participants

A total of 36 participants were initially recruited for this study. Six participants did not complete the intervention, resulting in a final sample of 30 older adults (mean age: 87.9 ± 5.6 years). Participants were divided into two groups: a control group performing conventional cycling without VR (Control, $n = 16$) and an intervention group performing cycling combined with VR (VR, $n = 14$).

Inclusion criteria included preserved mobility of the lower limbs, absence of joint pain, and intact lower-limb motor function, allowing participants to safely perform the cycling task. All participants were required to be able to understand the study procedures and provide written informed consent prior to participation.

Exclusion criteria comprised the presence of any implanted electronic device in the head, joint stiffness affecting the lower limbs, a history of vertigo or dizziness, and lower-limb surgery within the previous six months, as these conditions could interfere with safe participation in the exercise sessions or with the use of VR equipment.

2.2. Session Organization

This study followed a controlled and longitudinal intervention comparing conventional cycling exercise with cycling combined with VR. The intervention lasted six weeks, with three sessions per week, resulting in a total of 18 sessions per participant. Both groups followed the same cycling protocol, with the only difference being the use of VR in the intervention group.

The study was conducted at one of the residential care facilities of the Albertia Group, located in Majadahonda (Community of Madrid, Spain). The intervention protocol was jointly defined and agreed upon with the medical,

physiotherapy, and psychology teams of the residence to ensure safety, feasibility, and suitability for a very old population.

All exercise sessions were supervised by the technical team, with additional monitoring by the psychology team to ensure the well-being of the participants and appropriate engagement throughout the intervention.

2.3. Experimental Protocol

Each session began with a brief explanation of the protocol, and all participants provided written informed consent prior to participation. Participants in the VR group wore a head-mounted virtual reality device, while all participants were equipped with an inertial sensor placed on the thigh.

The VR intervention was implemented using *PedaleoVR*, a VR application developed by Werium Assistive Solutions for this project. *PedaleoVR* provides visual feedback on pedaling speed and includes a visual guide of the cadence that participants should maintain, with the objective of motivating users to sustain a constant exercise intensity during the cycling sessions.

The exercise task consisted of 20 minutes of cycling at a constant speed of 6 km/h. After each session, participants completed questionnaires assessing perceived effort, emotional state, and motivation. Functional mobility assessments were conducted at baseline and every two weeks throughout the intervention.

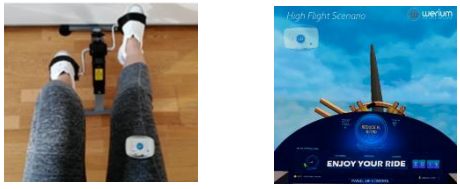


Figure 1. *Experimental setup and virtual reality interface.*

3. Data Analysis

3.1. Outcome Variables

The following outcome variables were analysed: physical performance, perceived effort, emotional state, and motivation.

Physical performance was assessed using the Timed Up and Go (TUG) test, a widely used and validated measure of functional mobility in older adults [4]. The test consists of standing up from a chair, walking a distance of three meters, turning around, walking back to the chair, and sitting down again. The total time required to complete the task was recorded in seconds, with lower values indicating better functional mobility.

Perceived effort was measured using the Borg scale, which provides a subjective assessment of exercise intensity and has been extensively used in clinical and exercise settings [5].

Emotional state was evaluated using the EVEA questionnaire, a validated instrument designed to assess transient emotional states. The questionnaire includes four subscales: anxiety, depression, hostility, and joy [6].

Motivation was assessed in the VR group using the Intrinsic Motivation Inventory (IMI), a multidimensional questionnaire commonly used to evaluate motivation in activity-based and experimental contexts [7]. The IMI subscales analysed in this study included interest, perceived usefulness, and perceived choice.

3.2. Data Processing and Statistical Analysis

Descriptive statistics were calculated for all variables, including mean, median, and standard deviation values for each group and assessment session. Data were visually analysed using line plots and boxplots to explore trends over time and variability between groups.

Differences between groups and changes over time were analysed using independent *t*-tests. Statistical significance was set at $p < 0.05$.

4. Results

4.1. Physical Performance

Physical performance was assessed using the TUG test. At baseline, the Control group showed substantially higher TUG times compared to the VR group (39.8 ± 35.5 s vs. 18.4 ± 10.1 s; Table 1). After the six-week intervention, mean TUG time in the Control group showed a slight decrease to 37.4 ± 36.0 s, whereas the VR group showed a small increase to 21.9 ± 17.9 s.

Despite these changes, the comparison between baseline and final assessments did not reveal statistically significant differences between groups (p = 0.173). A high degree of variability was observed, particularly in the Control group, as reflected by the wide dispersion of values and the presence of outliers (Figure 2).

Group	Basal(mean±SD)	Final(mean±SD)
C	39.8 ± 35.5	37.4 ± 36
VR	18.4 ± 10.1	21.9 ± 17.9

Table 1. *Mean ± SD values of TUG performance at baseline and final assessment (C: control group; VR: virtual reality group).*

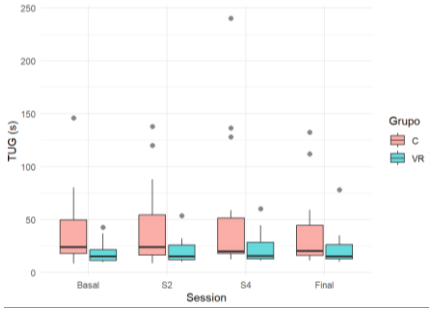


Figure 2. *Boxplot distribution of TUG performance by group and session.*

4.2. Perceived Effort (Borg Scale)

Perceived effort was evaluated using the Borg scale, ranging from 0 (no effort) to 5 (maximum effort). Baseline Borg scores were similar in both groups (Control: 3.5 ± 0.9; VR: 3.4 ± 1.4; Table 2). Over the intervention period,

perceived effort tended to decrease in both groups, with a more pronounced reduction observed in the VR group, particularly during the intermediate sessions, as illustrated in Figure 3.

Mean Borg scores decreased from 3.5 ± 0.9 to 3.3 ± 0.8 in the Control group and from 3.4 ± 1.4 to 2.8 ± 0.6 in the VR group. However, the comparison of changes between baseline and final assessments did not reveal statistically significant differences between groups (p = 0.377).

Group	Basal(mean±SD)	Final(mean±SD)
C	3.5 ± 0.9	3.3 ± 0.8
VR	3.4 ± 1.4	2.8 ± 0.6

Table 2. *Mean ± SD values of Borg perceived exertion at baseline and final assessment (C: control group; VR: virtual reality group).*

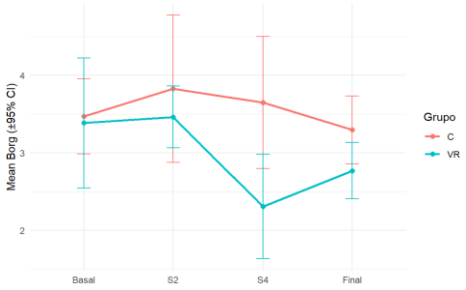


Figure 3. *Borg perceived exertion over time by group.*

4.3. Emotional State (EVEA)

Emotional state was assessed using the EVEA questionnaire, with subscales scored on a 0-10 scale. Anxiety, depression, and hostility scores remained low throughout the intervention in both groups, with no statistically significant differences observed between groups (Figure 4).

In contrast, joy scores were higher than the other EVEA subscales in both groups. Mean joy scores in the Control group remained stable from baseline to final assessment (6.1 ± 2.4 to 6.2 ± 2.5), while consistently higher values were observed in the VR group (7.0 ± 2.0 to 7.0 ± 2.3; Table 3; Figure 4). However, the comparison between groups did not reveal statistically significant differences.

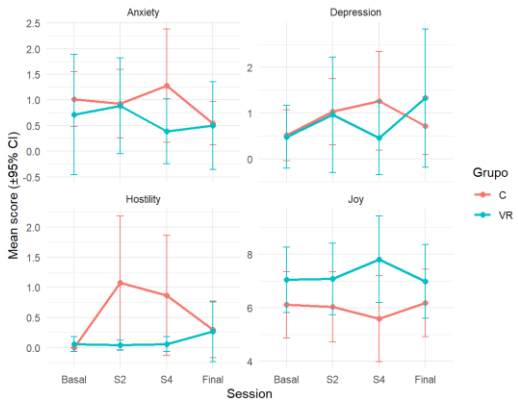


Figure 4. *EVEA subscale scores over time by group.*

Group	Basal(mean±SD)	Final(mean±SD)
C	6.1 ± 2.4	6.2 ± 2.5
VR	7.0 ± 2.0	7.0 ± 2.3

Table 3. *Mean ± SD values of EVEA joy at baseline and final assessment (C: control group; VR: virtual reality group).*

4.4. Motivation (IMI)

Motivation was assessed in the VR group using the IMI, with subscale scores ranging from 0 to 10. Interest, perceived usefulness, and perceived choice showed moderate values throughout the intervention period. From baseline to final assessment, mean scores for interest decreased from 6.03 ± 1.19 to 4.75 ± 2.54, and perceived usefulness decreased from 6.33 ± 0.99 to 4.65 ± 3.05. Perceived choice showed a smaller change, decreasing from 5.20 ± 1.06 to 4.93 ± 0.37 (Table 4; Figure 5).

Overall, despite the slight decreases observed in some IMI subscales, motivation levels remained within a moderate range throughout the virtual reality assisted cycling program.

Group	Basal(mean±SD)	Final(mean±SD)
Interest	6.03 ± 1.19	4.75 ± 2.54
Perceived usefulness	6.33 ± 0.99	4.65 ± 3.05
Perceived choice	5.20 ± 1.06	4.93 ± 0.37

Table 4. *Mean ± SD values of IMI subscales at baseline and final assessment.*

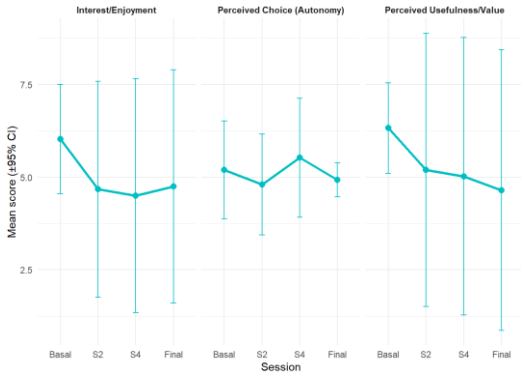


Figure 5. *IMI subscale scores over time in the VR group.*

5. Discussion

This study examined the effects of a six-week cycling program performed with and without VR in a group of very old adults, considering physical performance, perceived effort, emotional state, and motivation. Although most outcomes did not reach statistical significance, the results show several consistent patterns that support the potential benefits of VR as a supportive tool in exercise programs for older populations.

In terms of physical performance, no significant differences were found between the VR and Control groups in TUG results. This finding should be interpreted with caution, as participants in the VR group already showed better mobility at baseline, leaving limited room for improvement compared to the Control group. Importantly, performance in the VR group remained stable throughout the intervention, with less variability between participants. This reduced variability may be related to the immersive nature of the virtual environment, which could help participants remain more focused on the task, reduce perceived fatigue, or distract attention away from physical discomfort during exercise. Overall, these findings suggest that VR may help maintain functional performance in a more consistent way over time.

Perceived effort showed a relevant and favorable trend for the VR group. Although both groups reported lower effort levels as the program progressed, the reduction was more pronounced and more consistent in participants using VR. This finding is especially important in very old adults, since perceived effort strongly influences exercise tolerance and adherence to training. By providing an immersive environment, VR may help shift attention away from physical fatigue, making exercise feel less demanding and easier to sustain.

Emotional outcomes further support the positive role of VR. Anxiety, depression, and hostility remained very low in both groups throughout the intervention, indicating that the exercise program was emotionally safe. Participants in the VR group showed more stable emotional responses across sessions, with fewer fluctuations. In addition, joy scores tended to be higher in the VR group, suggesting that VR made the exercise experience more pleasant and enjoyable, even though these differences were not statistically significant.

Motivation results reinforce these findings. Motivation was assessed only in the VR group using the IMI. Interest, perceived usefulness, and perceived choice showed moderate values throughout the intervention, indicating that participants remained engaged and willing to continue the activity. The slight decrease observed in some IMI subscales from baseline to final assessment may reflect a natural adaptation to the repeated task over time or cumulative fatigue, rather than a loss of engagement, as motivation levels remained within a moderate range throughout the intervention. Maintaining motivation over six weeks in a very old population is a relevant outcome, as lack of motivation is a common limitation of traditional exercise programs.

The absence of statistically significant differences can be explained by several factors. The intervention duration may have been too short to produce measurable changes in physical and emotional outcomes in very old adults. In addition, the sample size was limited, and inter-individual variability, particularly in the Control group, may have contributed to reduced statistical power. Finally, participants showed low emotional symptoms and relatively good baseline functional status, which limited the magnitude of potential improvements.

Despite these limitations, the consistent patterns observed across perceived effort, emotional responses, and motivation suggest that VR offers meaningful benefits beyond physical performance alone. VR may improve the overall exercise experience by making it more comfortable, engaging, and emotionally positive for very old adults. Future studies including larger samples and longer intervention periods are needed to confirm these findings and to better determine the long-term value of VR-based exercise programs in this population.

6. Conclusion

The results of this study indicate that cycling with VR is well tolerated and provides a more positive exercise experience in very old adults. Although no statistically significant improvements in physical performance were observed, participants using VR tended to report lower perceived effort and more positive emotional responses during exercise. In addition, motivation was maintained throughout the intervention period.

From a practical perspective, these findings suggest that VR may help reduce perceived barriers to exercise and support continued participation in physical activity among very old adults. Overall, VR may represent a valuable supportive tool to enhance the exercise experience and promote active ageing in this population, particularly when the goal is to support comfort, enjoyment, and adherence to physical exercise.

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References

[1] J. Lee, S. Kim, and H. Park, “Development of a 360° virtual reality cycling system for older adults,” *BMC Geriatrics*, vol. 21, no. 1, pp. 1–10, 2021.

[2] A. Tegegne et al., “Acceptability of cycling with virtual reality among older adults living independently,” *Journal of Aging and Physical Activity*, 2025, ahead of print.

[3] J. Campo-Prieto et al., “Exergames for older adults: A randomized clinical trial,” *Sensors*, vol. 22, no. 18, p. 6742, 2022.

[4] D. Podsiadlo and S. Richardson, “The Timed ‘Up & Go’: A test of basic functional mobility for frail elderly persons,” *Journal of the American Geriatrics Society*, vol. 39, no. 2, pp. 142–148, 1991.

[5] G. Borg, “Psychophysical bases of perceived exertion,” *Medicine & Science in Sports & Exercise*, vol. 14, no. 5, pp. 377–381, 1982.

[6] J. Sanz, “Un instrumento para evaluar los estados emocionales transitorios: la Escala de Valoración del Estado de Ánimo (EVEA),” *Revista de Psicopatología y Psicología Clínica*, vol. 6, no. 2, pp. 93–109, 2001.

[7] E. McAuley, T. Duncan, and V. V. Tammen, “Psychometric properties of the Intrinsic Motivation Inventory in a competitive sport setting,” *Research Quarterly for Exercise and Sport*, vol. 60, no. 1, pp. 48–58, 1989.